

ARYAN PRAHRAJ

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SUMMARY

Software Engineer with 4 years of experience designing and developing scalable enterprise applications using Java, Spring Boot, and microservices architecture. Proficient in building secure RESTful APIs, cloud-native applications on AWS, and end-to-end CI/CD pipelines with modern DevOps practices. Experienced in backend development, distributed systems, relational and NoSQL databases, and responsive Angular applications. Strong background in financial services, application security, and Agile software delivery, with a passion for building reliable, high-performance solutions that address complex business and technical challenges.

CORE SKILLS

Programming Languages: Java (8/11/17), Python, JavaScript, TypeScript, SQL, PL/SQL, C++, Go

Frameworks & Libraries: Spring Boot, Spring MVC, Spring Security, Spring Cloud, Hibernate, JPA, Angular (14/17), React.js, Node.js, Express.js, Flask, RESTful APIs/Web Services, TensorFlow, Pandas, NumPy

Microservices & Distributed Systems: Microservices Architecture, Apache Kafka, API Gateway, Eureka Service Discovery, Resilience4j

Cloud & DevOps: AWS (EC2, ECS, EKS, Elastic Beanstalk, Lambda, S3, RDS, CloudWatch, VPC), Docker, Kubernetes, Jenkins, Maven, Terraform, Docker Compose, Git, CI/CD

Databases: Oracle, PostgreSQL, MySQL, MongoDB, Redis, DynamoDB, Cassandra, Elasticsearch, SQLite

Testing & Developer Tools: JUnit, Mockito, Postman, Swagger/OpenAPI, SonarQube, GitHub, GitLab, Bitbucket, Jira, Confluence, Slack, Visual Studio, Kubernetes Dashboard, ELK Stack

Methodologies: Agile (Scrum), SDLC, Test-Driven Development (TDD), Object-Oriented Programming (OOP), Design Patterns

PROFESSIONAL EXPERIENCE

Software Engineer | JPMorgan Chase & Co., Jersey City, NJ

Oct 2025 – Present

- Developed core backend services for a real-time credit card transaction processing platform using Java 17 and Spring Boot 3, supporting high-volume authorization, settlement, and fraud-screening workflows across a distributed microservices architecture.
- Implemented an event-driven architecture with Apache Kafka to stream transaction and settlement events between authorization, fraud detection, and ledger services, reducing processing latency and improving system scalability.
- Designed and secured RESTful APIs for internal card management and digital banking applications by implementing request validation, API versioning, and OAuth2/JWT authentication to ensure secure and reliable service integration.
- Containerized microservices with Docker and deployed them on Amazon EKS, configuring horizontal pod autoscaling, rolling updates, and health checks to maintain 99.9% platform availability during peak transaction volumes.
- Developed Angular modules for an internal transaction-monitoring dashboard, enabling fraud analysts to monitor real-time transaction activity, review alerts, and investigate suspicious payment events more efficiently.
- Optimized PostgreSQL database performance by refining schemas, indexing strategies, and query execution plans for transaction and ledger data, significantly improving response times for high-volume reporting and reconciliation workloads.
- Automated application delivery using Jenkins and AWS CodePipeline, building CI/CD pipelines that streamlined build, testing, and deployment processes across development, QA, and staging environments, accelerating release cycles.
- Strengthened platform reliability by integrating Resilience4j circuit breakers and centralized application logging with the ELK Stack, improving fault tolerance, accelerating production issue detection, and reducing mean time to resolution (MTTR).

Software Engineering Intern | Pitney Bowes, Shelton, CT

Jul 2025 – Oct 2025

- Remediated 100+ high-severity security vulnerabilities across Spring Boot and WildFly microservices by upgrading critical dependencies, including Jackson, SnakeYAML, and BouncyCastle, strengthening application security and improving IAM compliance by 30%.
- Modernized the build ecosystem by implementing Maven parent POMs, migrating legacy libraries, and removing deprecated components, simplifying dependency management and improving long-term maintainability.
- Containerized and deployed 10+ Java microservices on AWS ECS, EKS, and Elastic Beanstalk, implementing load balancing, auto scaling, and environment-specific configurations to achieve 99% service availability while reducing deployment time by 25%.
- Built and enhanced secure RESTful APIs using Spring Boot, Hibernate JPA, and Java, improving application performance, maintainability, and integration across distributed microservices.

- Streamlined CI/CD workflows using Jenkins, GitLab, and automated testing, reducing build failures by 40%, accelerating release cycles by 35%, and lowering manual validation effort by 15% through continuous integration and deployment automation.

Software Development Engineer | Cognizant Technology Solutions, India

Jun 2021 – Jul 2024

- Contributed to developing loan underwriting, eligibility, and approval modules using Java 8, Spring Boot, and Spring MVC, supporting end-to-end processing of retail and commercial loan applications.
- Designed and implemented RESTful APIs to integrate with external credit bureau and third-party services, automating credit score retrieval, risk evaluation, and applicant verification while improving processing efficiency.
- Built responsive Angular user interfaces for the loan officer portal, delivering features for document management, application tracking, and real-time status updates to enhance user experience and operational productivity.
- Developed and optimized Oracle PL/SQL stored procedures, functions, and complex SQL queries, improving high-volume loan data processing and reporting performance across transactional databases.
- Implemented Spring Security with role-based access control (RBAC) and authentication mechanisms to secure sensitive customer and financial data in accordance with enterprise security and data governance standards.
- Automated build and deployment workflows using Jenkins, Git, and Bitbucket, streamlining CI/CD processes across development and UAT environments while improving release reliability.
- Created comprehensive unit and integration tests using JUnit and Mockito, increasing code quality, improving test coverage, and reducing defects identified during regression testing.
- Collaborated within Agile Scrum teams alongside business analysts, QA engineers, and UI/UX designers to translate business requirements into scalable software solutions, participating in sprint planning, code reviews, and iterative feature delivery.

PROJECTS

VirtualPal – Multi-Persona Conversational AI System with Persona-Locked Inference

- Developed VirtualPal, a multi-persona conversational AI platform that delivers secure, context-aware interactions through persona-specific inference pipelines. Built a server-side orchestration layer to preserve conversation context, enforce session isolation, and maintain consistent persona behavior. Implemented a rule-driven persona framework with safety controls to minimize persona drift and support reliable, real-time AI conversations across multiple use cases.

Guard My Bills – Machine Learning-Driven Fraud Detection & Transaction Risk Intelligence Platform

- Designed and implemented Guard My Bills, a privacy-first machine learning platform for real-time financial fraud detection. Developed an anomaly detection model leveraging over 30 behavioral and geospatial features to assess transaction risk, providing accurate fraud detection, low-latency inference, and scalable processing of high-volume transactions.

SheetEasy – Cloud-Native Intelligent Spreadsheet Platform

- Developed SheetEasy, a cloud-native spreadsheet platform using Next.js, React, and TypeScript with support for multi-sheet state management, high-performance computation, offline synchronization, and real-time autosave. Integrated an AI copilot capable of interpreting natural language instructions to perform data transformations, complex calculations, automated visualizations, and PNG exports, delivering an intuitive, AI-powered data analysis experience.

FreeFlix – Full-Stack Movie Discovery Platform

- Engineered FreeFlix, a full-stack movie discovery platform using React, Node.js, Express, MongoDB, Firebase Authentication, and TMDB/OMDb APIs. Built secure backend services for user authentication, personalized watchlists, viewing history, and real-time movie recommendations across multiple languages and mood-based categories, delivering a scalable and responsive cloud-hosted application.

EDUCATION

Master of Science in Computer Science

Pace University, New York, New York, USA

Bachelor of Technology in Computer Science and Engineering

SOA University, Bhubaneswar, Odisha, India

CERTIFICATIONS

- AWS Certified Developer – Associate